```python
class Book:
    def __init__(self, title, author, isbn):
        self.title = title
        self.author = author
        self.isbn = isbn

    def __str__(self): #since we will  use composition, we need a string for the function to return when used in user functions
        return f"'{self.title}' by {self.author} (ISBN: {self.isbn})"

class User:
    def __init__(self, name):
        self.name = name
        self.borrowed_books = []

    def borrow_book(self, book):
        if book not in self.borrowed_books:
            self.borrowed_books.append(book)
            print(f"{self.name} has borrowed {book.title}.")
        else:
            print(f"{self.name} already borrowed {book.title}.")

    def return_book(self, book):
        if book in self.borrowed_books:
            self.borrowed_books.remove(book)
            print(f"{self.name} has returned {book.title}.")
        else:
            print(f"{self.name} does not have {book.title}.")

    def list_borrowed_books(self):
        if self.borrowed_books:
            print(f"{self.name} has borrowed:")
            for book in self.borrowed_books:
                print(f"{book}")
        else:
            print(f"{self.name} has not borrowed any books.")

Book1 = Book("Dork Diaries","Rachel Renee Russell", 9781416980063)
Book2 = Book("Diary of a Wimpy Kid","Rachel Renee Russell", 9788498672220)

Person1 = User("Sandina")
Person1.borrow_book(Book1) #using composition
Person1.borrow_book(Book2)
Person1.list_borrowed_books()
Person1.return_book(Book1)
Person1.list_borrowed_books()
```

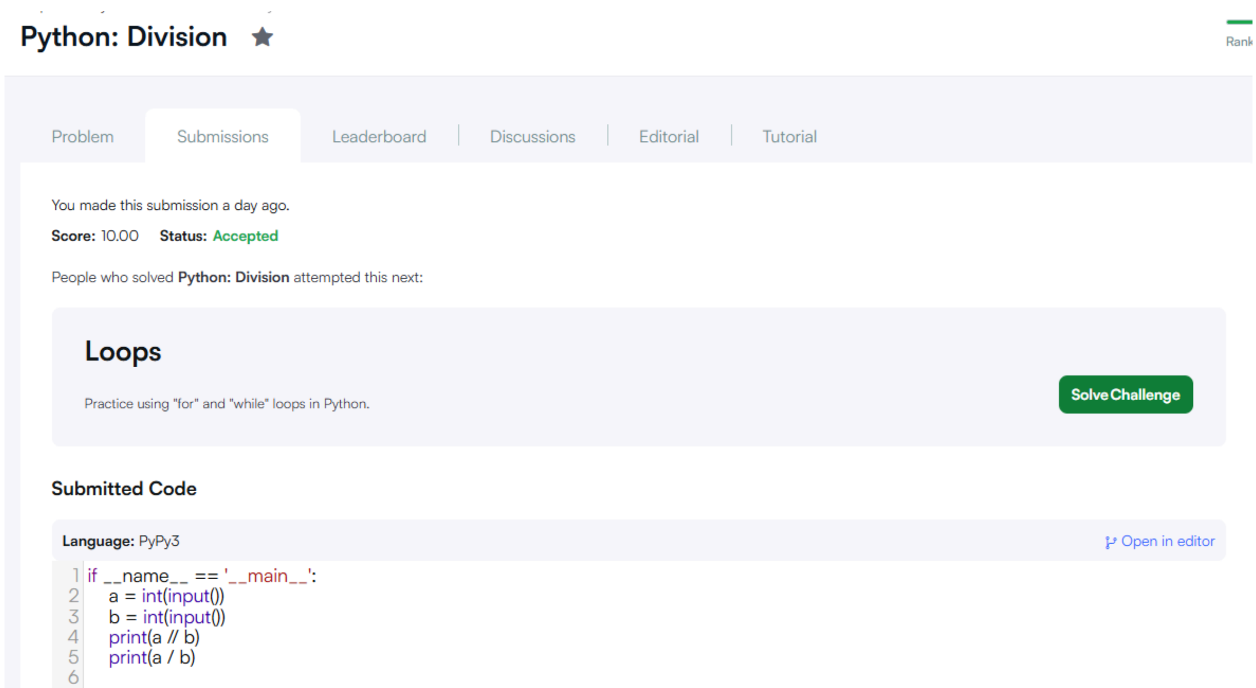

```python
Person1 = User("Sandina")
Person1.borrow_book(Book1) #using composition
Person1.borrow_book(Book2)
Person1.list_borrowed_books()
Person1.return_book(Book1)
Person1.list_borrowed_books()
```


```
...
Sandina has borrowed Dork Diaries.
Sandina has borrowed Diary of a Wimpy Kid.
Sandina has borrowed:
'Dork Diaries' by Rachel Renee Russell (ISBN: 9781416980063)
'Diary of a Wimpy Kid' by Rachel Renee Russell (ISBN: 9788498672220)
Sandina has returned Dork Diaries.
Sandina has borrowed:
'Diary of a Wimpy Kid' by Rachel Renee Russell (ISBN: 9788498672220)
```

Code Explanation: To make the BOOK class, i have to define all the arguments as what they represented using self. I also used a self string so when I would later utilize whatever book i created in the user section, it would have something to print out. This is cause I KNEW i was going to use composite classes. Otherwise it will cause some type of mishap. Next, for the user class, i made a list of books, which are books they borrowed, and made functions that allow you to borrow books (append to list of return books) or return them (remove them from the list). For both classes I had a function to allow me to see at a glance what the status is. For Book it was the __str__(self) and for User it was list_borrowed_books().

EASY (5)

1



2.

Say "Hello, World!" With Python ★

Ra

Problem | Submissions | Leaderboard | Discussions | Editorial | Tutorial

You made this submission a day ago.

Score: 5.00 **Status:** Accepted

People who solved **Say "Hello, World!" With Python** attempted this next:

Loops

Practice using "for" and "while" loops in Python.

Solve Challenge

Submitted Code

Language: PyPy3 Open in editor

```
if __name__ == '__main__':
    print("Hello, World!")

```

3.

Print Function ★

Rank:

Problem | Submissions | Leaderboard | Discussions | Editorial | Tutorial

You made this submission 15 hours ago.

Score: 20.00 **Status:** Accepted

People who solved **Print Function** attempted this next:

Validating and Parsing Email Addresses

Print valid email addresses according to the constraints.

Solve Challenge

Submitted Code

Language: Python 3 Open in editor

```
if __name__ == '__main__':
    n = int(input())
    for i in range(1,n+1):
        print(i,end="")

```

4.

You made this submission a day ago.

Score: 10.00 **Status:** Accepted

People who solved **Python If-Else** attempted this next:

Loops

Practice using "for" and "while" loops in Python.

Solve Challenge

Submitted Code

Language: PyPy3 Open in editor

```
#!/bin/python3

import math
import os
import random
import re
import sys



if __name__ == '__main__':
    n = int(input().strip())
    print("Weird" if n % 2 != 0 or (n <= 20 and n >= 6 and n % 2 == 0) else "Not Weird")

```

5.

Problem | Submissions | Leaderboard | Discussions | Editorial | Tutorial

You made this submission a day ago.

Score: 10.00 **Status:** Accepted

People who solved **Arithmetic Operators** attempted this next:

Loops

Practice using "for" and "while" loops in Python.

Solve Challenge

Submitted Code

Language: PyPy3 Open in editor

```
if __name__ == '__main__':
    a = int(input())
    b = int(input())
    print(a + b)
    print(a - b)
    print(a * b)

```

MEDIUM(3)

1.

You made this submission 14 hours ago.

Score: 40.00 **Status:** Accepted

People who solved **ginortS** attempted this next:

Re.start() & Re.end()

Find the indices of the start and end of the substring matched by the group.

Solve Challenge

Submitted Code

Language: Python 3

Open in editor

```
odd = []
even = []

for i in string:
    if i.isupper():
        upper.append(i)
    elif i.islower():
        lower.append(i)
    elif i.isnumeric():
        if int(i) % 2 == 0:
            even.append(i)
        else:
            odd.append(i)

print("".join(sorted(lower)) +  "".join(sorted(upper)) + "".join(sorted(odd)) + "".join(sorted(even)))
```

2.

Write a function ★

Rank: 72

Problem | Submissions | Leaderboard | Discussions | Editorial

You made this submission a day ago.

Score: 10.00 **Status:** Accepted

People who solved **Write a function** attempted this next:

Default Arguments

Debug a function which uses a default value for one of its arguments.

Solve Challenge

Submitted Code

Language: PyPy3 Open in editor

```
def is_leap(year):
    leap = False
    if 1900 <= year <= 10**5:
        if year % 4 == 0 and (year % 100 != 0 or year % 400 == 0):
            leap = True
        else:
            leap = False
    else:
        pass
    return leap
```

3. The minion game

```python

    #Kevin Point Management
    kevin_points = 0

    #Kevin Point Control
    for i in range(len(string)):
        if string[i] in vowels:
            kevin_points += len(string) - i
        else:
            stuart_points += len(string) - i

    #Comparing points
    if kevin_points > stuart_points:
        print(f"Kevin {kevin_points}")
    elif kevin_points < stuart_points:
        print(f"Stuart {stuart_points}")
    else:
        print("Draw:")


if __name__ == '__main__':···
```

Upload Code as File

Test against custom input

Congratulations!

You have passed the sample test cases. Click the submit button to run your code against all the test cases.

HARD(2)

Maximize It! ★

Problem | Submissions | Leaderboard | Discussions | Editorial

You made this submission 14 hours ago.

Score: 50.00 **Status:** Accepted

People who solved **Maximize It!** attempted this next:

String Validators

Identify the presence of alphanumeric characters, alphabetical characters, digits, lowercase and uppercase characters in a string.

Solve Challenge

Submitted Code

Language: Python 3 Open in editor

```
#    length, modulo = int(k[0]), int(k[1])
#    masterlist = []
#    for i in range(length):
#        inputs = list(map(int,input.split()))
#        masterlist.append(inputs[:])
#    func = map(lambda x: sum((i**2 for i in masterlist)%2), product(*masterlist))
#    print(max(func))
from itertools import product
listnum, mod = list(map(int,input().split()))
masterlist = []
for i in range(listnum):
    sublist = list(map(int,input().split()))[1:]
    masterlist.append(sublist)
func = map(lambda x: sum(i*i for i in x) % mod, product(*masterlist))
print(max(func))
```

2.

Problem

Submissions

Leaderboard

Discussions

Editorial

$0 < N, M < 100$

Note: A 0 score will be awarded for using 'if' conditions in your code.

Output Format

Print the decoded matrix script.

Sample Input 0

```
7 3
Tsi
h%x
i #
sM 
$a 
#t%
ir!
```

Sample Output 0

```
This is Matrix#  %!
```

Explanation 0

The decoded script is:

```
This$#is% Matrix#  %!
```

Neo replaces the symbols or spaces between two alphanumeric characters with a single space ' ' for better readability.

So, the final decoded script is:

```
This is Matrix#  %!
```

```python
# Input rows and columns
first_multiple_input = input().rstrip().split()
n = int(first_multiple_input[0])
m = int(first_multiple_input[1])

# Read the matrix
matrix = [input() for i in range(n)]

# Read column-wise
decoded = ''.join(''.join(row[i] for row in matrix) for i in range(m))
decoded_script = re.sub(r'(?<=\w)([^\w]+)(?=\w)', ' ', decoded)

print(decoded_script)
```

Upload Code as File

Test against custom input

Congratulations!